

COMP2019 Systems Programming 1

Tut/Prac Two

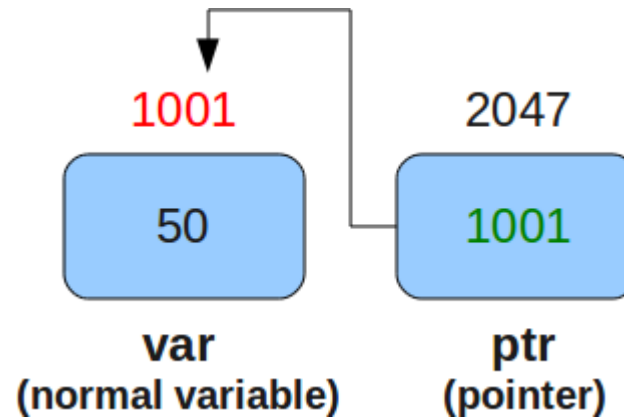
Pointers

Are an effective way to represent linked data structures.

- Provide a higher level of abstraction than parallel arrays.
E.g. binary tree $\leftrightarrow$ pointer to node
- Allow access to dynamic (or run-time) storage for objects of indefinite size, or transient existence.
- In C a function can be regarded as a pointer constant and it is possible to define a pointer to function variable and assign different functions to it.

Pointers

- Address 1001 stores 50, address 2047 stores 1001



Pointers and Arrays

- A pointer type variable is declared using the `*` operator, e.g.
`int *s;`
which says that `s` is a pointer to `int`. This type expression is also pronounced int star.
- In C an array is also represented as a pointer constant.
So a declaration such as `int q[12];` is taken by the compiler to mean that `q` is a constant pointer to the first of twelve `ints`; referred to as `q[0]`, `q[1]`, ..., `q[11]`.
- The subscripting operator `[ ]` is defined as follows:
$$q[k] \Leftrightarrow *(q + k)$$

so in particular `q[0]` simply means `*q`, i.e. contents of `q`.

Structures in C

Declarations struct

```
struct playertype
{
    char name[30];
    int score;
};
```

```
struct playertype
{
    char name[30];
    int score;
} players[10];
```

```
struct playertype players[10];
```